

BIOMARKER KITCHEN

12H | 7-10 DAYS | 48-72H | BASELINE

REINFARCTION DETECTOR

SECONDARY PREVENTION SHELF

INDEFINITELY

25% ↓ RISK

CONDITION LVEF ≤40% OR HF

ACE

LVEF ≤40% + HF + HF OR DM

MINERAL

ELEVATED BANANA (DIFFERENTIAL K⁺ ≥5.0)

ELECTROLYTE STATION

POTASSIUM 4.5

MAGNESIUM 2.0

MEASURE + CORRECT ON ADMISSION

RETURN TO ACTIVITY TIMELINE

3-5 DISCHARGE GATE

2 MONTHS?

3-5 METs = 2 FLIGHTS OF STAIRS

~1 WEEK SEXUAL ACTIVITY RESUMPTION

2-4 WEEKS MARKS RETURN TO WORK

1. Biomarker kitchen — troponin

WHAT YOU SEE

Left 'BIOMARKER KITCHEN' counter with a steaming cooked STEAK on a plate (the muscle marker) — the slab of meat = troponin released from injured cardiac muscle.

WHAT IT ENCODES

Cardiac troponin (I or T) is the most sensitive and specific marker of myocardial necrosis and the preferred test for diagnosing MI. It rises ~3–12h after injury, peaks around 24h, and stays elevated for 7–10 days (troponin T) / 5–7 days (troponin I). Diagnosis requires a rise and/or fall with at least one value above the 99th percentile, plus ischemic context. Mnemonic: 'troponin: up by 3–12h, peaks at a day, lingers a week-plus.'

BOARD TRAP

Board says: 'Troponin normalizes within 24h of an MI.' WRONG. Discriminator: troponin stays elevated for 7–10 days; that is why it cannot reliably detect early REINFARCTION — for a suspected re-MI within days, use CK-MB, which has already returned toward baseline.

2. Troponin 7–10 days

WHAT YOU SEE

Timeline '12H ... 7–10 DAYS ... BASELINE'

WHAT IT ENCODES

Because troponin remains elevated for 7–10 days, it is excellent for diagnosing an MI even days after presentation, but POOR for detecting reinfarction during that window (a persistently high level can't distinguish old from new injury). In the reinfarction window, look for a new RISE in CK-MB or a re-rise (≥20%) in troponin from a prior plateau. Mnemonic: 'troponin lingers a week+ → blind to re-MI in that window.'

BOARD TRAP

Board says: 'A patient with chest pain on day 4 post-MI and a still-elevated troponin has definitely reinfarcted.' WRONG. Discriminator: troponin is expected to remain elevated 7–10 days; to confirm reinfarction you need a NEW rise (CK-MB, or ≥20% re-rise in troponin), not just a persistently high value.

3. CK-MB reinfarction detector

WHAT YOU SEE

Plated dish 'REINFARCTION DETECTOR', '48–72H'

WHAT IT ENCODES

CK-MB rises ~3–6h, peaks at ~24h, and returns to baseline by 48–72h. This short, predictable course makes CK-MB the marker of choice to detect EARLY REINFARCTION: after the first CK-MB normalizes, a new rise indicates fresh myocardial necrosis. Its lower specificity (also in skeletal muscle) is the trade-off. Mnemonic: 'CK-MB clears in 2–3 days → catches the second hit.'

BOARD TRAP

Board says: 'Use troponin rather than CK-MB to detect reinfarction a few days after an MI.' WRONG. Discriminator: troponin is still elevated and uninformative in that window; CK-MB has normalized, so its re-rise specifically flags REINFARCTION — CK-MB is the right tool here.

4. Onset 6–12H

WHAT YOU SEE

Bed '6–12H' on timeline

WHAT IT ENCODES

Biomarkers and ECG findings EVOLVE over the first 6–12h, so SERIAL testing is mandatory — a single early measurement can miss an evolving MI. Obtain troponin at presentation and repeat at 3–6h (and again if needed); follow serial ECGs. The classic 0/1h or 0/3h high-sensitivity troponin protocols exist precisely because of this time-dependence. Mnemonic: 'one normal troponin is not enough — recheck over 6–12h.'

BOARD TRAP

Board says: 'A single normal troponin at presentation rules out MI.' WRONG. Discriminator: troponin may not rise until 3–12h after symptom onset; a single early normal value does NOT exclude MI — serial troponins (and ECGs) are required.

5. Secondary prevention shelf

WHAT YOU SEE

Right wall 'SECONDARY PREVENTION SHELF' — a lit display case lining up all the discharge meds (referee, statin, ACE/BB/aldosterone cards, knight) that every post-MI patient leaves on for life.

WHAT IT ENCODES

Every post-MI patient leaves on a lifelong secondary-prevention bundle: antiplatelet therapy (aspirin indefinitely + P2Y12 inhibitor), high-intensity statin, beta-blocker, and ACE inhibitor/ARB (especially with reduced EF/HF/DM/HTN), plus an aldosterone antagonist if EF ≤40% with HF or DM. Add risk-factor control (BP, glucose, smoking cessation) and cardiac rehab. Mnemonic: 'ABCDE — Aspirin/Antiplatelet, Beta-blocker, Cholesterol (statin)/Cardiac rehab, Diet/Diabetes, Exercise + ACEi.'

BOARD TRAP

Board says: 'Secondary-prevention medications can be stopped once the patient is discharged and feels well.' WRONG. Discriminator: these agents (aspirin, statin, beta-blocker, ACEi) are continued LONG-TERM/lifelong to reduce reinfarction and death — feeling well is not a reason to stop them.

6. Aspirin indefinitely

WHAT YOU SEE

Referee 'INDEFINITELY' on shelf

WHAT IT ENCODES

Aspirin (81 mg daily) is continued INDEFINITELY after MI. Dual antiplatelet therapy (DAPT) adds a P2Y12 inhibitor — clopidogrel 75 mg, ticagrelor 90 mg BID, or prasugrel 10 mg — for at least 12 months after ACS (shorter if high bleeding risk, longer if high ischemic/low bleeding risk). Antiplatelet therapy gives roughly a 25% relative risk reduction in recurrent vascular events. Mnemonic: 'aspirin forever, the partner (P2Y12) for about a year.'

BOARD TRAP

Board says: 'Stop aspirin at 12 months after the MI.' WRONG. Discriminator: it is the P2Y12 INHIBITOR that is typically stopped around 12 months; ASPIRIN is continued INDEFINITELY/lifelong. Stopping aspirin removes the lifelong antiplatelet backbone.

7. Statin high-intensity

WHAT YOU SEE

On the shelf, an 'ATORVA 40-80' bottle wearing a gold star — the high-dose statin every MI patient gets regardless of LDL, starred as mandatory.

WHAT IT ENCODES

All post-MI patients get a HIGH-INTENSITY statin (atorvastatin 40–80 mg or rosuvastatin 20–40 mg) regardless of baseline LDL, for both LDL lowering and pleiotropic plaque-stabilizing effects. Goal LDL <70 mg/dL (many guidelines now target <55 mg/dL for very-high-risk patients); add ezetimibe ± a PCSK9 inhibitor if goals are not met. Mnemonic: 'every MI patient gets a high-intensity statin — no LDL excuse.'

BOARD TRAP

Board says: 'Start a statin only if the LDL is elevated.' WRONG. Discriminator: post-MI, high-intensity statin is indicated for ALL patients regardless of baseline LDL — the benefit is independent of starting cholesterol and includes plaque stabilization.

8. ACEi if EF ≤40% / HF / DM

WHAT YOU SEE

Shelf 'ACEi', 'LVEF ≤40%', 'HF', 'DM'

WHAT IT ENCODES

Start an ACE inhibitor (or ARB if intolerant) within 24h in stable patients, with the greatest benefit when there is LVEF ≤40%, heart failure, anterior MI, hypertension, diabetes, or CKD — it limits adverse remodeling and reduces mortality. Titrate to target while monitoring K⁺ and creatinine. Add an aldosterone antagonist (spironolactone/eplerenone) if EF ≤40% with HF or DM. Mnemonic: 'ACEi for the weak/diabetic/failing heart — blocks remodeling.'

BOARD TRAP

Board says: 'ACE inhibitors are contraindicated after MI.' WRONG. Discriminator: ACEi/ARB are guideline-recommended post-MI (class I when EF ≤40%/HF/DM/HTN) and reduce mortality — true contraindications are limited (hyperkalemia, bilateral RAS, pregnancy, angioedema, hypotension), not the MI itself.

9. Beta-blocker shelf

WHAT YOU SEE

On the prevention shelf, the 'BB' card paired with 'LVEF ≤40%' — beta-blocker continued long-term, biggest benefit when the EF is low.

WHAT IT ENCODES

Oral beta-blockers (metoprolol, carvedilol, bisoprolol) are started within 24h in stable patients and continued LONG-TERM; they reduce ischemia, reinfarction, arrhythmia, and mortality, with the largest benefit when EF is reduced. AVOID IV beta-blockers / early initiation in patients with signs of heart failure, hypotension, cardiogenic shock, or high heart-block risk. Mnemonic: 'beta-blocker long-term — but not while the patient is in shock or florid HF.'

BOARD TRAP

Board says: 'Beta-blockers are only needed for the first week after an MI.' WRONG. Discriminator: beta-blockers are continued LONG-TERM (especially with reduced EF) for ongoing mortality benefit — they are not a short-course drug. (The acute caution is avoiding them in decompensated HF/shock, not stopping them at one week.)

10. Electrolyte station K & Mg

WHAT YOU SEE

Right 'ELECTROLYTE STATION', potassium/magnesium

WHAT IT ENCODES

Maintain potassium >4.0 mEq/L and magnesium >2.0 mg/dL after MI to lower the risk of ventricular arrhythmias (VT/VF) in the irritable peri-infarct period. Hypokalemia and hypomagnesemia are correctable arrhythmia triggers, so aggressive repletion is standard. Mnemonic: 'keep K above 4, Mg above 2 — quiet the irritable myocardium.'

BOARD TRAP

Board says: 'Electrolytes no longer matter once the artery is revascularized.' WRONG. Discriminator: even after successful PCI, hypokalemia/hypomagnesemia provoke peri-infarct ventricular arrhythmias — maintaining K⁺ >4.0 and Mg²⁺ >2.0 remains important for arrhythmia prevention.

11. Discharge gate ~24h

WHAT YOU SEE

'DISCHARGE GATE', '24H' chair

WHAT IT ENCODES

Modern post-STEMI care favors EARLY mobilization (out of bed within ~24h) and short stays: an uncomplicated, successfully reperfused STEMI patient is often dischargeable in a few days (sometimes 48–72h for low-risk PCI patients). Early ambulation prevents deconditioning and venous thromboembolism while the secondary-prevention bundle is started. Mnemonic: 'reperfused and uncomplicated → up early, home in days.'

BOARD TRAP

Board says: 'Patients need strict bed rest for weeks after a STEMI.' WRONG. Discriminator: prolonged bed rest is outdated and harmful (VTE, deconditioning); current practice is EARLY mobilization and short hospitalization for uncomplicated, reperfused MI.

12. Sexual activity ~1 week

WHAT YOU SEE

Heart 'SEXUAL ACTIVITY RESUMPTION ~1 WEEK', 2 METs

WHAT IT ENCODES

Sexual activity can usually be resumed ~1 week after an uncomplicated MI in patients who are asymptomatic and can achieve modest exertion (~3–5 METs, e.g., climbing two flights of stairs without symptoms). Counsel that PDE5 inhibitors are CONTRAINDICATED with nitrates (risk of severe hypotension). Mnemonic: 'two flights of stairs symptom-free → back to the bedroom (~1 week).'

BOARD TRAP

Board says: 'Sexual activity is prohibited for 6 months after an MI.' WRONG. Discriminator: after an uncomplicated MI, sexual activity may resume in about 1 week once the patient tolerates ~3–5 METs — a blanket 6-month ban is incorrect (the real caveat is no nitrates with PDE5 inhibitors).

13. Return to work 2–4 weeks

WHAT YOU SEE

'2–4 WEEKS' return-to-work marker

WHAT IT ENCODES

Most patients with an uncomplicated MI can return to work in about 2–4 weeks, individualized by job physical demands, LV function, and results of risk stratification (e.g., exercise testing). Physically strenuous jobs may require longer or a graded return; sedentary work can resume sooner. Cardiac rehab supports a safe return. Mnemonic: 'uncomplicated MI → back to work in 2–4 weeks.'

BOARD TRAP

Board says: 'All MI patients must stay off work for 6 months.' WRONG. Discriminator: return to work after an uncomplicated MI is typically 2–4 weeks, individualized to job demands and functional capacity — a universal 6-month restriction is incorrect.

14. 3–5 METs / 2 flights of stairs

WHAT YOU SEE

Stairs '3–5 METs = 2 FLIGHTS OF STAIRS'

WHAT IT ENCODES

Functional capacity is gauged in METs: climbing two flights of stairs (~3–5 METs) without symptoms is the practical benchmark that the heart can tolerate ordinary activities, including sexual activity and many jobs. Formal exercise/stress testing before discharge or early after can quantify capacity and risk-stratify. Cardiac rehabilitation is a class I recommendation to improve fitness, mortality, and quality of life. Mnemonic: 'two flights of stairs ≈ 3–5 METs ≈ green light for daily activity.'

BOARD TRAP

Board says: 'No functional assessment is needed before clearing a post-MI patient for activity.' WRONG. Discriminator: activity clearance is based on demonstrated functional capacity (e.g., ~3–5 METs / two flights of stairs, often via exercise testing) and cardiac rehab — not given blindly without assessment.
